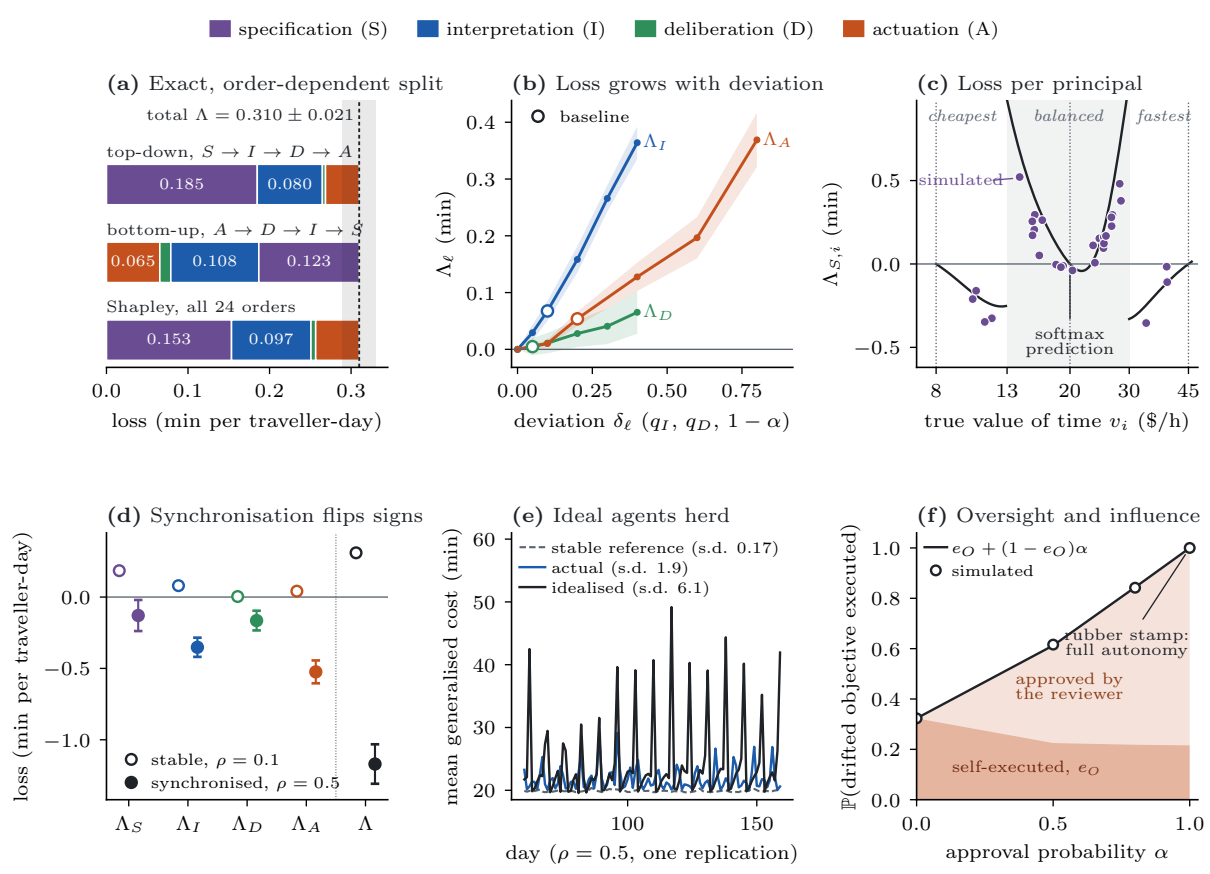

specification (S)
interpretation (I)
deliberation (D)
actuation (A)
(a) Exact, order-dependent split
total Λ = 0.310 ± 0.021
top-down, S → I → D → A
0.185
0.080
bottom-up, A → D → I → S
0.065
0.108
0.123
Shapley, all 24 orders
0.153
0.097
0.0
0.1
0.2
0.3
loss (min per traveller-day)
(b) Loss grows with deviation
baseline
Λ_I
Λ_A
Λ_D
Λ_ℓ (min)
0.00
0.25
0.50
0.75
deviation δ_ℓ (q_I, q_D, 1 − α)
(c) Loss per principal
cheapest
balanced
fastest
simulated
softmax prediction
Λ_{S,i} (min)
8
13
20
30
45
true value of time v_i ($/h)
(d) Synchronisation flips signs
loss (min per traveller-day)
stable, ρ = 0.1
synchronised, ρ = 0.5
Λ_S
Λ_I
Λ_D
Λ_A
Λ
(e) Ideal agents herd
stable reference (s.d. 0.17)
actual (s.d. 1.9)
idealised (s.d. 6.1)
mean generalised cost (min)
day (ρ = 0.5, one replication)
(f) Oversight and influence
e_O + (1 − e_O)α
simulated
rubber stamp: full autonomy
approved by the reviewer
self-executed, e_O
ℙ(drifted objective executed)
approval probability α